vestigated numerically and experimentally. We have verified experimentally that as far as the impedance bandwidth of a λ/2 patch antenna is concerned, it is better to use a magneto-dielectric substrate as introduced here rather than a dielectric substrate. The measurements indicate that the antenna efficiency remains high when the proposed magneto-dielectric substrate is used as a filling material, thus validating the feasibility of the proposed design.

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COMPACT BROADBAND U-SLOT-LOADED RECTANGULAR MICROSTRIP ANTENNAS

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Received 7 March 2005

ABSTRACT: *A compact variation of a U-slot-loaded rectangular microstrip antenna—a half U-slot (L-shaped slot) loaded rectangular microstrip antenna—is proposed. The bandwidth of a shorted compact square microstrip antenna is increased by cutting a quarter-wavelength step U-slot.* © 2005 Wiley Periodicals, Inc. Microwave Opt Technol Lett 46: 556–559, 2005; Published online in Wiley InterScience (www.interscience.wiley.com). DOI 10.1002/mop.21049

Key words: *U-slot rectangular microstrip antenna; half U-slot rectangular microstrip antenna; stepped U-slot rectangular microstrip antenna; compact microstrip antenna*

1. INTRODUCTION

A compact microstrip antenna (MSA) is obtained by using either a higher dielectric constant ε_r substrate or a shorting post, or by cutting a slot [1, 2]. The bandwidth (BW) is increased by using either thicker substrates with lower ε_r or multiresonator gap-coupled and stacked configurations [3]. However, these BW improvement techniques increase the antenna size or its thickness. Broadband MSAs using a resonant slot cut inside the patch increases the BW without increasing its volume [4]. The slot adds another resonant mode near the TM_{10} resonance frequency of unslotted rectangular MSAs (RMSAs), thereby increasing the BW.

A broadband configuration using a U-slot cut inside the RMSA, a circular MSA, and an equilateral triangular MSA have been reported [4–6]. In this paper, a U-slot RMSA is discussed and its even-mode equivalent, that is, a compact half U-slot (L-shaped slot) RMSA (whose size is reduced by half) is proposed. A variation of the U-slot, the stepped U-slot RMSA, is proposed and the BW of a compact shorted square MSA (SMSA) is increased by cutting a half stepped U-slot. All the MSAs have been initially optimized using the IE3D software [7], followed by experimental verification. For the simulation of all the MSAs, an infinite ground plane has been assumed. However, for the experiments, the size of ground plane is taken to be more than six times the substrate thickness in all directions with respect to the patch dimensions so as to reduce the effect of finite ground plane [3].

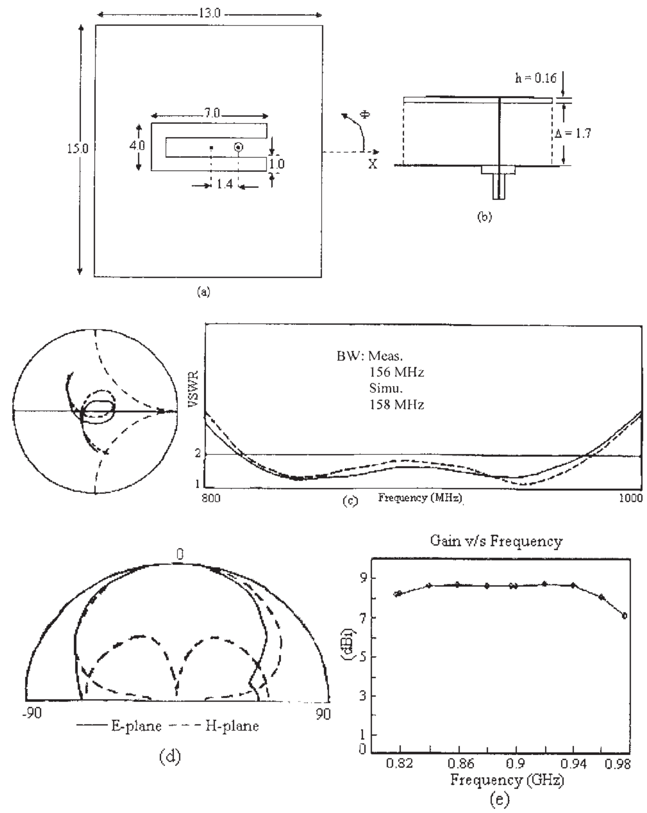

Figure 1 Suspended U-slot RMSA; (a) top and (b) side views; (c) input impedance and VSWR plots (—— measured, – – – simulated); (d) simulated radiation pattern at 894 MHz and its (e) gain variation with frequency

2. U-SLOT RMSA

A U-slot RMSA etched on a glass epoxy substrate ($\varepsilon_r = 4.3$, $h = 0.16$ cm, and $\tan\delta = 0.02$), is suspended over the ground plane with an air gap Δ of 1.7 cm, as shown in Figures 1(a) and 1(b). The total thickness h_t is 1.859 cm ($0.06\lambda_0$). The RMSA was designed to operate at around 900 MHz. The resonant U-slot is located symmetrically with respect to the feed-point axis. The broadband response of the U-slot RMSA depends upon the length and width of the slot, substrate thickness, and the feed-point location. The slot dimensions are selected such that its resonance frequency is close to that of the patch frequency. Due to the broader BW of the U-slot RMSA, the patch width is selected such that the TM_{02} mode lies outside the desired band [8].

By optimizing the various parameters, a broadband response is obtained. The slot is nearly $\lambda/2$ in length and its inner length governs the resonant characteristics. The measured and simulated input impedance and VSWR plots are shown in Figure 1(c). The measured BW is from 816 to 972 MHz (156 MHz, 17.4%) whereas the simulated BW is 158 MHz (17.6%). The simulated radiation pattern at 894 MHz, as shown in Figure 1(d), is in the broadside direction and the cross-polar level is less than 15 dB, as compared to the co-polar level. The gain is more than 8 dBi over the operating BW and it decreases towards the higher band frequencies. Towards the higher frequencies, the slot mode is dominant, which creates an asymmetrical field distribution around the patch. This increases the cross-polar radiation, which reduces the gain. The surface-current distribution generated using Ensemble software [9] at the center frequency f_C and the two band edge frequencies f_L and f_H are shown in Figures 2(a)–2(c), respectively. At f_L, the surface currents show similar variation as that in the unslotted RMSA in the region outside the U-slot, whereas at f_H the surface current has a half-wavelength variation along the inner-slot length.

3. HALF U-SLOT-LOADED RMSA

By employing the even-mode symmetry of the U-slot RMSA with respect to the feed-point axis and using only half of the configuration, a half U-slot RMSA is obtained, as shown in Figures 3(a) and 3(b). The width is decreased by half, which increases the radiating resistance and the mutual coupling between the slot and the patch mode. This increases the loop size, which lies outside the VSWR = 2 circle. Thus, h_t is increased to 2.159 cm, which increases the fringing field, thereby reducing the radiating resistance and the loop size (which lies inside the VSWR = 2 circle). The optimized configuration is shown in Figure 3(c). The measured BW is 130 MHz (14.6%), whereas the simulated BW is 128 MHz (14.2%), as shown in Figure 4(a). The radiation pattern at 901 MHz is shown in Figure 4(b). The maximum is in the broadside direction and the cross-polar level is less than 12 dB, as compared to that of the co-polar level. The gain is more than 6 dBi over the BW with a peak gain of 8 dBi. Thus, by using the symmetry of U-slot RMSA, 50% reduction in the antenna size is obtained.

4. STEPPED U-SLOT-LOADED RMSA

The shorted compact MSAs have a smaller BW. A resonant U-slot will be too large to be placed inside the compact MSA. Hence, a stepped U-slot which is obtained by redistributing the horizontal arm length of the U-slot into two steps is proposed, as shown in Figure 5(a). The resonant length is along the inner stepped length. For optimized dimensions, the measured and simulated input impedance and VSWR plots are shown in Figure 5(b). The measured BW is 126 MHz (14.3%), whereas the simulated BW is 133 MHz (15.1%). The radiation pattern is in the broadside direction with a cross-polar level of less than 15 dB, as compared to that of the co-polar level. The gain is more than 8 dBi over the operating BW. Compared with the U-slot RMSA, the stepped U-slot RMSA has a slightly reduced BW and gain.

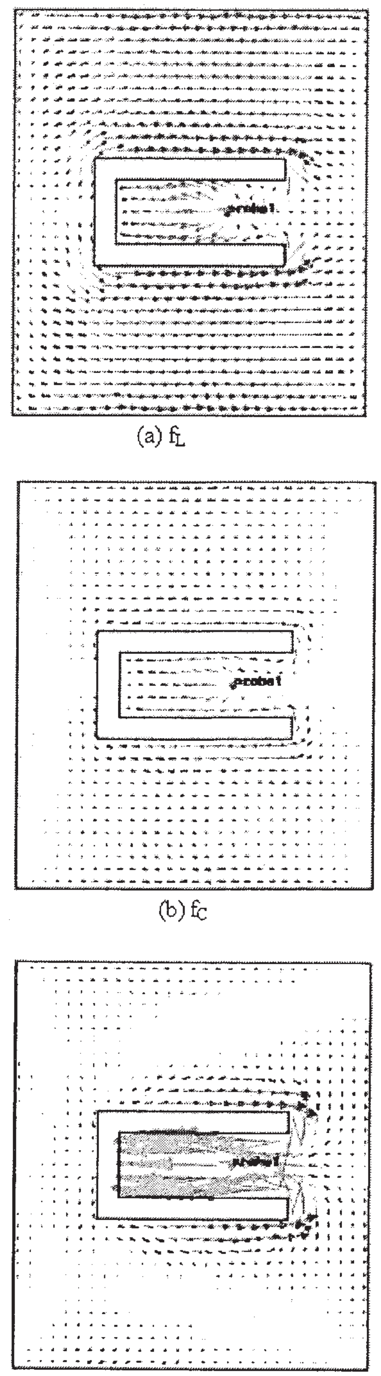

Figure 2 Surface current distribution for the U-slot RMSA at (a) f_L, (b) f_C, and (c) f_H

5. COMPACT BROADBAND SLOTTED SMSAS

The BW of the shorted-plate SMSA is improved by cutting a quarter-wavelength stepped U-slot. The dimensions of the slot are chosen in such a way that its resonance frequency is near the resonance frequency of the shorted patch, which increases the overall BW. The optimized dimensions of the antenna with h = 2.8 cm on foam substrate ($\varepsilon_r = 1.1$, $\tan\delta = 0.00001$) are shown in Figure 6(a). The larger substrate thickness is required in order to reduce the coupling between the slot and patch mode. The measured and simulated input impedance and VSWR plots are shown in Figure 6(b). The measured BW is 204 MHz (23.5%), whereas

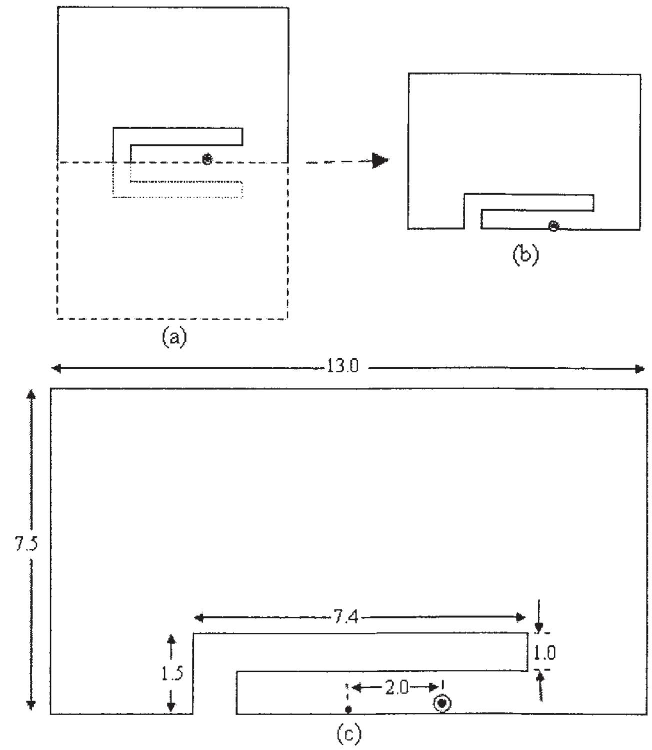

Figure 3 (a) U-slot RMSA and its (b) even-mode symmetry and (c) optimized half U-slot RMSA

the simulated BW is 199 MHz (22.7%). The radiation pattern at 877 MHz in the E- and H-planes is shown in Figure 6(c), which shows higher cross-polarization levels. The antenna gain is more than 3 dBi over the operating BW, which is smaller than that of the RMSA due to the decreased aperture area. Although the cross-polar levels are higher due to the shorted patch and increased probe radiation, they may be useful in the mobile-communication environment.

A very compact MSA can be realized by using a single shorting post, but it will have narrow BW. A quarter-wavelength stepped resonant slot is used to increase the BW of the edge center-shorted

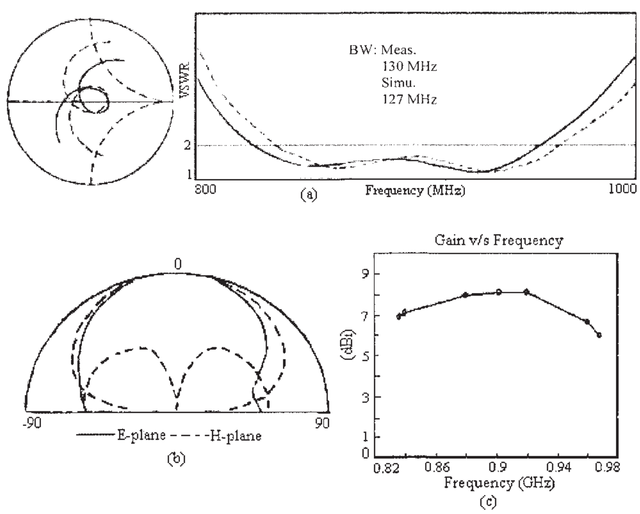

Figure 4 (a) Input impedance and VSWR plots of the half U-slot RMSA (—— measured, – – – simulated); (b) simulated radiation pattern at 901 MHz and its (c) gain variation with frequency

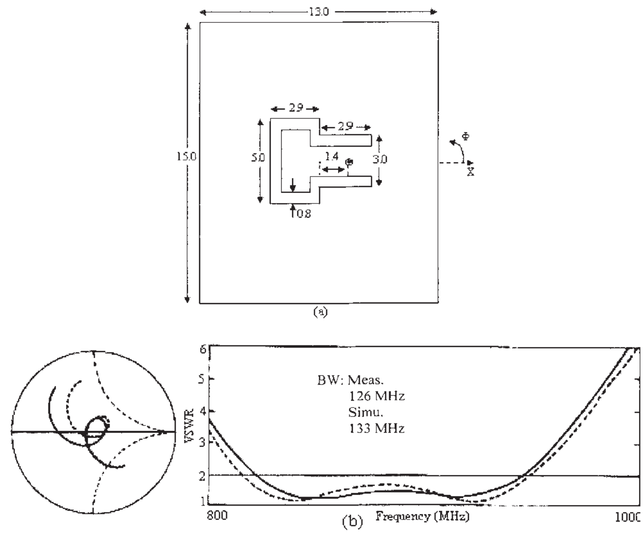

Figure 5 (a) RMSA with a stepped U-slot and its (b) input impedance and VSWR plots (—— measured, – – – simulated)

SMSA, as shown in Figure 7(a). A probe and shorting post of diameters 0.12 and 0.1 cm, respectively, are used. A MSA fabricated on a glass epoxy substrate was used in a three layer suspended configuration with $\Delta = 2.6$ cm as shown in Figure 7(b), giving $h_t = 2.918$ cm ($0.086\lambda_0$). The measured and simulated input impedance and VSWR plots for the optimized parameters are shown in Figure 7(c). The measured BW is 153 MHz (17.3%), whereas the simulated BW is 165 MHz (18.5%). The simulated radiation pattern at 892 MHz is shown in Figure 7(d). The maxi-

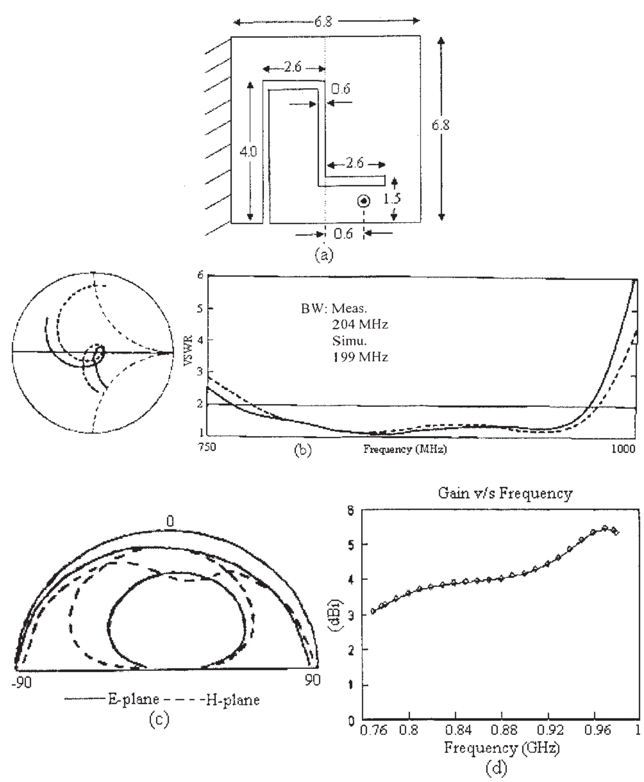

Figure 6 (a) Half stepped U-slot shorted plate SMSA; (b) input impedance and VSWR plots (—— measured, – – – simulated); (c) simulated radiation pattern at 877 MHz and its (d) gain variation with frequency

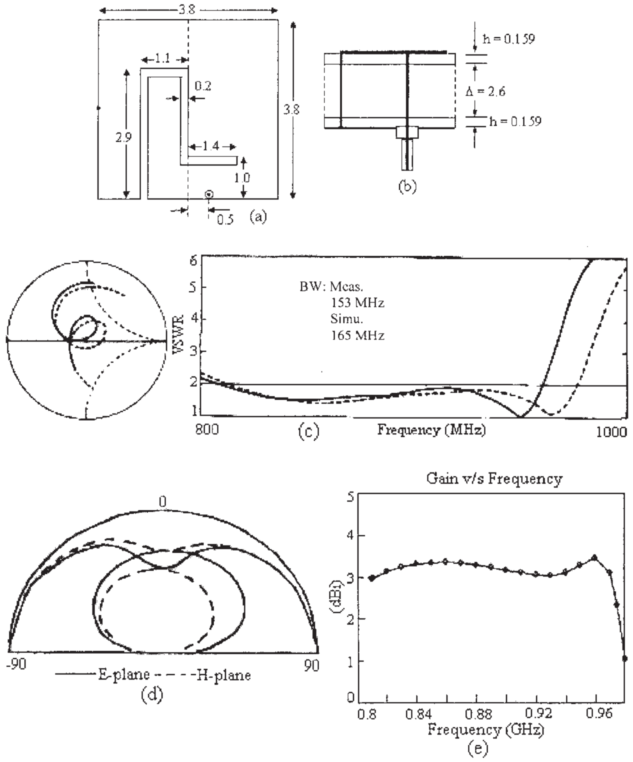

Figure 7 Edge center shorted SMSA with a stepped slot: (a) top and (b) side views; (c) input impedance and VSWR plots (—— measured, – – – simulated); (d) simulated radiation pattern at 892 MHz and its (e) gain variation with frequency

mum is in the end-fire direction with higher cross-polar level and a gain of 3 dBi is observed over the VSWR BW. The area of the proposed compact MSA is 15 cm^2, which is only 6% of the corresponding unslotted RMSA, which results in a better BW but a reduced gain.

6. CONCLUSION

A compact variation of the RMSA with a half U-slot, which reduces the antenna area by half, has been proposed. The BW enhancement of compact shorted SMSAs was obtained by cutting a quarter-wavelength stepped U-slot, which gives a BW of more than 20%.

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DESIGN OF A THINNED MICROSTRIP-ANTENNA REFLECTARRAY USING A GENETIC ALGORITHM

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Received 30 March 2005

ABSTRACT: *This paper presents the design of a thinned reflectarray. The phase-shifting elements are slotted rectangular microstrip antennas with via connection to the ground plane. They are disposed in a regular grid. The thinning procedure is used to determine which elements may be removed, according to the desired radiation pattern, without significant change in directivity. For this approach, a genetic algorithm is proposed.* © 2005 Wiley Periodicals, Inc. Microwave Opt Technol Lett 46: 559–562, 2005; Published online in Wiley InterScience (www.interscience.wiley.com). DOI 10.1002/mop.21050

Key words: *slotted microstrip antenna; via; genetic algorithm; antenna arrays; reflectarrays*

1. INTRODUCTION

The reflectarray concept is not new, but the rapid development of microstrip-antenna technology, coupled with the need for low cost, high gain, and aesthetically pleasing antennas for commercial applications, has led to the use of microstrip elements in a variety of reflectarray configurations [1–5].

In order to focus an incoming wave towards a receiver, the planar microstrip reflectarray uses a large array of phase-shifting elements. Each element radiates a part of the incoming wave with a particular phase-shift, in such a way that all radiations are in-phase in the desired direction. According to the incoming wave and the desired reflected wave, an individual phase shift for each element is determined. Several techniques can be used to involve this phase shift, for example, a small change of element resonant frequency. In this paper, we use slotted rectangular microstrip antennas with via connection to the ground plane. This technique was presented previously so as to achieve the required phases in order to realize beam-steering for a conventional reflectarray [6], and the first part of this paper deals with the design of the phase-shifting elements. However, once the main beam direction has been obtained, it may be necessary to adjust the shape of radiation pattern, for instance, in order to minimize the side-lobe level. This is addressed in the second part of this paper. A synthesis method is used to thin the array. It is based on microstrip antenna-array synthesis with the binary-feeding law [8]. In such a classical array, the synthesis determines which elements must be fed or not fed. In the case of a reflectarray, it determines whether an element must be present or not. To solve this problem, a genetic-algorithm approach is adopted [9]. An experimental reflectarray is realized and measured co- and cross-polarization radiation patterns are presented.

2. DESIGN OF PHASE-SHIFTER ELEMENTS

The reflection phase is analyzed using an equivalent unit-cell model waveguide approach (WGA) [7] with Ansoft high-fre-